

H2BENCH · REQUIREMENTS

H2Bench requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/h2bench		
Source	docs/03-requirements.md @ 8f87588		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 status from HBN-CAL-001; R14 scope states that H2Guard is costed in its own project (HBN-DDR-001)
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with first-order status
0.1	2026-09-25	Amish Chadha	Initial scaffold

H2Bench requirements

These requirements were checked by calculation at TRL 3 in HBN-CAL-001 (docs/04-calcs/01-sizing.md). Targets are unchanged from v0.2 except R14, whose scope now states that H2Guard is costed in its own project and shipped with the bench (adopted for TRL 3 under Amish's 2026-09-25 instruction, open for his review; HBN-DDR-001 item 6). Nine requirements are met on paper, **two are not met (R9 and R14)**, four are at risk (R5, R8, R10 and R13) and one (R16) cannot be verified until hardware exists. "Met" means met by calculation or design review, not by test.

Table 1. Requirements and status at TRL 3.

ID	Requirement	Target	Status (HBN-CAL-001)	Verification (TRL 3 or later)
R1	Measure the efficiency of each stage	Electrolyzer, storage, fuel cell and round trip, each to within ± 5 % relative	Met: hydrogen per fill ± 1.5 %, worst stage ± 1.8 %	Uncertainty calculation (HBN-CAL-001 section 6)
R2	Complete a full cycle in one lesson	Fill and discharge within 60 min of run time	Met: 17.7 min fill plus 29.0 min discharge, 46.7 min	Energy and flow calculation
R3	Electrolyzer power	100 W or less electrical input	Met: 70.2 W (79 W with aged cells)	Calculation; stack datasheet later
R4	Fuel cell power	50 W or less; 10 W or more net to run a visible load	Met: 10.8 W net, a 0.8 W margin (9.8 W if cells fall to 0.55 V)	Calculation; stack datasheet later
R5	Limit the hydrogen inventory	10 L or less at 20 °C and 101.3 kPa	At risk: 8.5 L at 300 kPa gauge, 9.6 L at the relief setting, 10.5 L at full relief lift on a 15 °C day	Gas law calculation
R6	Limit storage pressure	Working pressure 300 kPa gauge or less; relief at 350 kPa gauge or less; vessel rated 1 MPa or more	Met by design: hoop stress 18 MPa at 1 MPa	Component ratings
R7	No flammable mixture in the room from a total release	Full inventory mixed into a 30 m ³ room below 25 % of the lower flammable limit	Met: 0.032 % by volume, 0.8 % of LFL	Calculation
R8	Interlock shuts down on a leak	H2Guard alarm cuts electrolyzer power and closes the tank solenoid within 2 s; fan runs	At risk: 1.1 s from H2Guard's TRL 2 figures, but H2Guard is at TRL 2 and its valve is 24 V against H2Bench's 12 V solenoid	H2Guard design review
R9	Hydrogen quality at the fuel cell	Purity as the stack requires (the 12 W class specifies 99.995 % or better)	Not met: water is removed by the drier, but oxygen crossover is unknown and there is no deoxidizer	Supplier data, later gas test
R10	Electrolyzer tolerates tank back-pressure	H2 side rated to 300 kPa gauge or more over the O2 side	At risk: the stack must hold 307 kPa while filling and 357 kPa at the relief setting; no rating in hand	Supplier data
R11	Water quality	Conductivity 1 μ S/cm or less at the stack inlet, shown to students	Met by design (mixed-bed resin and conductivity check)	Datasheet
R12	Log data students can analyze	Voltage, current, power, tank pressure and temperature at 1 Hz, saved as CSV	Met by design: 2,804 rows, about 224 kB per cycle	Firmware sketch review
R13	Fit an existing lab bench	Footprint 1,000 x 500 mm or less; height 800 mm or less; mass 25 kg or less	At risk: 900 x 450 x 755 mm, but 25.2 kg (22.2 kg without the bench supply)	Parametric model and mass estimate
R14	Low cost and buildable	Parts cost \$450 or less (<code>budget_usd</code>), excluding H2Guard, which is costed in its own project and	Not met: \$865 (\$800 without the bench supply). A budget of about	Priced BOM

ID	Requirement	Target	Status (HBN-CAL-001)	Verification (TRL 3 or later)
		shipped with the bench; hand tools only; no welding	\$850 is recommended, awaiting Amish (HBN-DDR-001 item 9)	
R15	Guard people from pressure parts	Tank inside a guard; relief and vent piped to the hood; no pressurized fitting that can be opened without a tool	Met by design: 18 mm guard clearance; lines to the canopy in the model	Design review
R16	Set up and shut down quickly	Setup 15 min or less; tank vented and bench safe in 5 min or less	Not verifiable at TRL 3: the vent takes 3.0 min at the needle valve setting; setup needs a walk-through with hardware	Walk-through when hardware exists

■ Assumptions

- Electrolyzer: 4 cells at about 1.95 V and 9 A, Faraday efficiency 95 %; hydrogen higher heating value (HHV) 285.8 kJ/mol.
- Tank: 2.000 L internal plus 0.15 L of lines at tank pressure, filled from 70 to 300 kPa gauge (171 to 401 kPa absolute) at 20 °C; below 70 kPa gauge the regulator cannot hold the fuel cell supply. The full assumption list is in HBN-CAL-001 Table 1.
- Fuel cell: 13 cells at 0.60 V and 1.5 A, fuel utilization 95 % (the rest purged), about 0.9 W for fan and controls.
- Room: 30 m³, well mixed. The canopy hood exists because mixing near the leak is not instant; see HBN-PRC-001.
- Round-trip figures use the HHV basis. On a lower heating value (LHV) basis the hydrogen energy is about 15 % lower and the fuel cell efficiency correspondingly higher.